

Lab 27 Worksheet — Printer Fault Diagnosis from Print Defects

Name: _____ Date: ____

Exam objective: Diagnose printer faults by reading the physical defect on the page and mapping it to the failed component in the print process (Core 1 objective 5.6).

Record as you go

Step	What you did	What you observed
1	Record the diagnostic principle: the pattern of the defect identifies the component, and a repeating defect's spacing identifies which rotating part by its circumference.	
2	Diagnose vertical lines running down every page on a laser printer as a scratched imaging drum, and record that the fix is drum or cartridge replacement.	
3	Diagnose toner that smudges or wipes off the page as a failed fuser — the toner was placed correctly but never melted onto the paper.	
4	Diagnose repeated ghost or echo images as a drum that is not being cleaned between cycles, and record that replacing the toner cartridge usually resolves it.	
5	Diagnose faded print as low toner or ink, and record the field workaround of gently rocking a laser cartridge to redistribute remaining toner.	
6	Diagnose garbled or nonsense output as the wrong print driver language — a PostScript driver sending to a PCL-only printer, or a corrupt driver.	
7	Diagnose entirely blank or entirely black pages as a primary corona or charging failure, since the	

Step	What you did	What you observed
	drum is either never charged or never discharged.	
8	Diagnose paper jams by location: the pickup area points at worn pickup rollers or separation pads, the fuser area points at the fuser or wrong paper weight.	
9	Diagnose multi-page misfeeds as worn separation pads or paper that is damp, curled or of the wrong weight for the tray.	
10	Diagnose incorrect colour output on an inkjet by checking cartridge seating, checking for third-party ink, running head cleaning and running a colour calibration.	
11	Diagnose incorrect paper size or orientation as an application, driver or tray configuration mismatch rather than a hardware fault, and record all three places to check.	
12	Build the connectivity checklist for a printer that prints nothing at all: check the print queue for stalled jobs, restart the print spooler, verify network reachability and confirm the correct printer is selected.	

Verification

Success criterion: Your map covers all twelve defects with a named component and a corrective action; the fuser, drum and corona faults are correctly distinguished; and the connectivity checklist includes clearing the queue and restarting the spooler.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
